ECON-7102: Environmental Economics I

Class 17: The price of time and recreation demand

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Fall 2026

Today's agenda

1. Questions?
2. Today:
   - Rapid-fire pitches: abstract 4
   - Travel cost basics (P&R 17.1 and 17.4.2)
   - "Are environmental values cyclical?" (working paper)

Hotelling (1947), letter to the National Park Service

The University of North Carolina
Institute of Statistics
Chapel Hill

DEPARTMENT OF MATHEMATICAL STATISTICS

June 18, 1947

Mr. Newton B. Drury, Director
National Park Service
Department of the Interior
Washington 25, D. C.

Dear Mr. Drury:

After a letter from Mr. A. E. Demaray, and a conference with Dr. Roy A. Pruitt of the National Park Service, I am convinced that it is possible to set up appropriate measures for evaluating, with a reasonable degree of accuracy, the service of national parks to the public.

The development of criteria for evaluating benefits to the public has been a long-term interest of mine. Following the example set a hundred years ago by the French engineer, Jules Dupuit, who wrote formulae for the benefits of roads, bridges, and canals, I have worked out more general formulae for benefits from wider and more complicated classes of public services.

These formulae, of course, involve coefficients which must, in each case, be determined by factual statistical studies. The development of such studies I believe to be possible through several modes of attack which Dr. Pruitt and I discussed. One of these, of whose feasibility I am confident, and which might be pursued further, is as follows:

Let concentric zones be defined around each park so that the cost of travel to the park from all points in one of these zones is approximately constant. The persons entering the park in a year, or a suitably chosen sample of them, are to be listed according to the zone from which they come. The fact that they come means that they service of the park is at least worth the cost, and this cost can probably be estimated with fair accuracy. If we assume that the benefits are the same no matter what the distance, we have, for those living near the park, a consumers' surplus consisting of the differences in transportation costs. The comparison of the cost of coming from a zone with the number of people who do come from it, together with a count of the population of the zone, enables us to plot one point for each zone on a demand curve for the service of the park. By a judicious process of fitting it should be possible to get a good enough approximation to this demand curve to provide, through integration, a measure of the consumers surplus resulting from the availability of the park. It is this consumers surplus (calculated by the above process with deduction for the cost of operating the park) which measures the benefits to the public in the particular year. This, of course, might be capitalized to give a capital value for the park, or the annual measure of benefit might be compared directly with the estimated annual benefits on the hypothesis that the park area was used for some alternate purpose.

2 -- Mr. Newton B. Drury June 18, 1947

The problem of relations between different parks can be treated along the same lines, though in a slightly more complicated manner, provided people entering the park will be asked which other national parks they have visited that year. In place of a demand curve, we have as a result of such an inquiry, a set of demand functions. The consumer surplus still has a defining meaning, as I have shown in various published articles, and may be used to evaluate the benefits from the park system.

This approach through travel costs is one of several possible modes of attack on this problem. There are also others, which should be examined, though I think the method outlined above looks the most promising.

Very sincerely,
Harold Hotelling
Harold Hotelling

HH:a

Reprinted in National Park Service (1949), *The Economics of Public Recreation*.

The travel-cost model (P&R 17.1)

- Hotelling (1947) proposed pricing a recreation trip by its money and time cost of travel.
  - The first form of nonmarket valuation, applied to valulation of National Parks.
- Substituting the time constraint into the budget constraint:

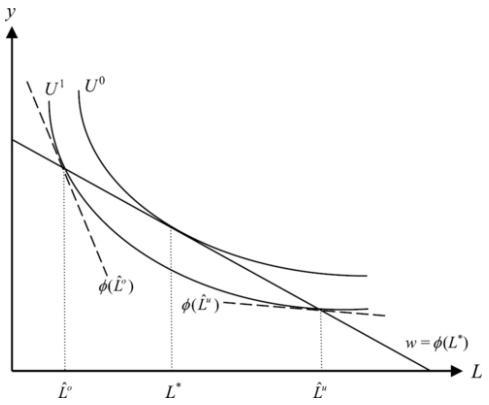
$$\max \; U(x, z, l, q) \quad \text{s.t.} \quad wT = z + w\,l + \underbrace{(c + w\,t)}_{p}\,x$$

  - x: trips, c: money cost, t: travel time, w: wage
  - $p = c + wt$: the "full price" of a trip
- First-order condition for an interior x:

$$\frac{\partial U/\partial x}{\partial U/\partial z} = c + w\,t$$

$\Longrightarrow$ **Time is priced at the wage.**

The price of time (P&R 17.1.2 and 17.4.2)



- If hours are not flexible, the two constraints do not combine. With multipliers λ (money) and μ (time):

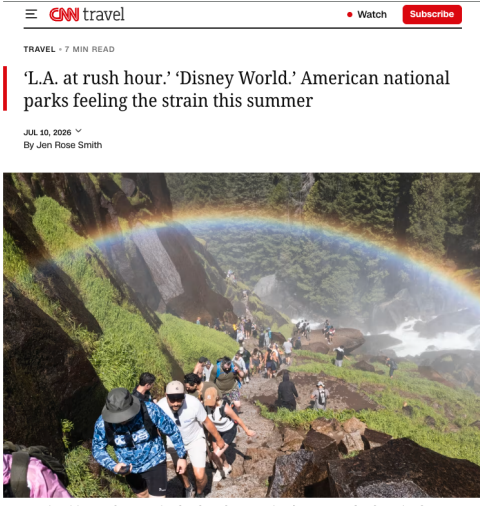
$$\frac{\partial U/\partial x}{\partial U/\partial z} = c + \underbrace{\frac{\mu}{\lambda}}_{\phi} t$$

- Over-employed: $\phi > w$. Under-employed: $\phi < w$.
- Common practice sets $\phi = w/3$ (Cesario, 1976; McConnell and Strand, 1981).
- **Q:** What fraction of the wage should we use? (P&R 17.2b)

λ and ϕ vary with the business cycle

- A travel-cost study assumes ϕ and estimates how trips respond to p, which depends on λ. Both are fixed at their data-year values.
- Neither is a preference parameter. In a downturn:
  - ϕ decreases: wages decrease and more people are under-employed.
  - λ increases: people consume less.
- **Q:** Are recreation values cyclical?

Motivation



CNN travel

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‘L.A. at rush hour.’ ‘Disney World.’ American national parks feeling the strain this summer

JUL 10, 2026

By Jen Rose Smith

Booming visitor numbers at national parks such as Yosemite, above, comes after the National ...

https://www.cnn.com/2026/07/10/travel/travel-national-parks-strain-summer-crowds

“In a year when many Americans are feeling financially strained and airfare prices remain high, it seems intuitive that national parks would rise in popularity. After all, a park vacation featuring hikes and picnics is a relatively affordable alternative to an international trip.”

Environmental values rest on two cyclical parameters

Environmental values are often inferred from time-allocation decisions.

- The opportunity cost of time $\omega = \mu/\lambda$ prices time inputs (Becker, 1965; DeSerpa, 1971).
- The marginal utility of income λ plays two roles:
  - An increase in λ makes demand steeper.
  - λ sets the exchange rate between welfare and dollars.
- Both vary with the business cycle.

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Benefit–cost analyses, damage assessments, and natural-capital accounts apply values dated by their studies' data years to damages and decisions dated in other years.

What I do

- I derive how demand, real surplus, and nominal surplus change over the business cycle.
- Empirically:
  - I estimate the cyclicality of National Park visitation by a proxy for travel time.
  - I illustrate the implications of cyclicality for published travel-cost values (RUVD, 1956–2023).
  - I rescale Deepwater Horizon damage assessments to a mid-cycle labor market.

What I do

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  - I rescale Deepwater Horizon damage assessments to a mid-cycle labor market.

Two primary conclusions:

1. Real and nominal surplus are cyclical, and can move in opposite directions, depending on the travel-time share.
2. Published travel-cost estimates are *incidentally acyclical* because three cyclical biases offset.

Both parameters are cyclical

Index the business cycle by the unemployment gap g ($g > 0$ is a downturn). The access price is $p^*(g) = c + \omega(g)\, t$.

In a downturn...

- The price of time decreases. With job loss, the average price of time decreases by more than the wage (Aguiar, Hurst, and Karabarbounis, 2013):

$$\Delta\bar{\omega} \approx \underbrace{e\,\Delta\omega^e}_{\text{wage margin}} + \underbrace{(\omega^e - \omega^n)\,\Delta e}_{\text{employment margin}}$$

- The marginal utility of income increases: $\Delta \ln \lambda = -\gamma\, \Delta \ln C$.

Over the Great Recession, the price of time decreased 7.5 to 10 percent and consumption decreased 3 percent. At $\gamma = 1.5$, λ increased 4.5 log points.

Demand is influenced by both cyclical parameters

$$x(g) = \exp\left(\alpha + \delta q - \beta(g)\, p^*(g)\right), \qquad \beta(g) = b\,\lambda(g)$$

- A money price deters trips in proportion to λ (Small and Rosen, 1981; Hanemann, 1984).
  - Standard practice estimates β and reports per-trip WTP as $V = 1/\beta$. Marginal WTP for quality is δ/β.
  - Dividing by $\beta = b\lambda$ converts utility into dollars at the data-year λ. I treat b as fixed and λ as cyclical.
- In a downturn, λ increases and the demand curve becomes steeper.
- In a downturn, ω decreases and the access price decreases in proportion to the travel-time share $s = \omega t/(c + \omega t)$.

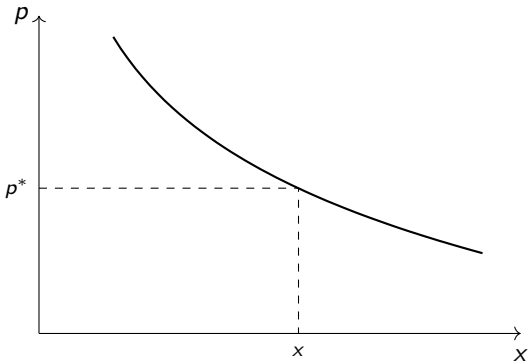
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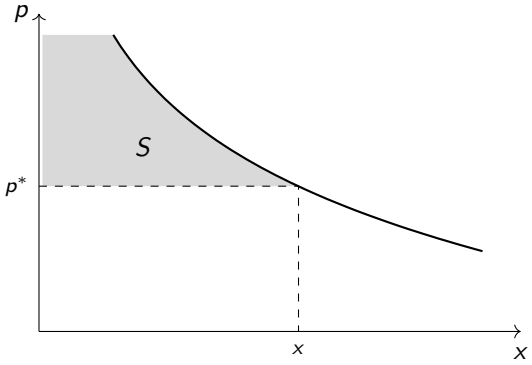
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The two cyclical changes affect demand in opposite directions, and the travel-time share decides which one dominates.

Real and nominal surplus

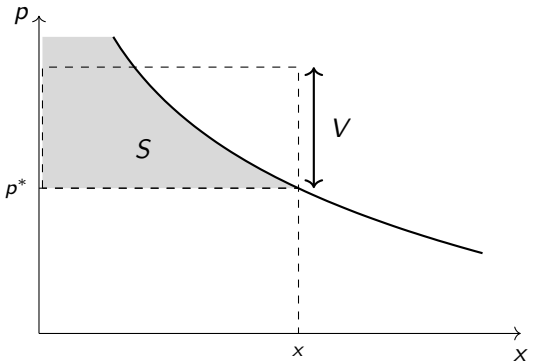


Real and nominal surplus



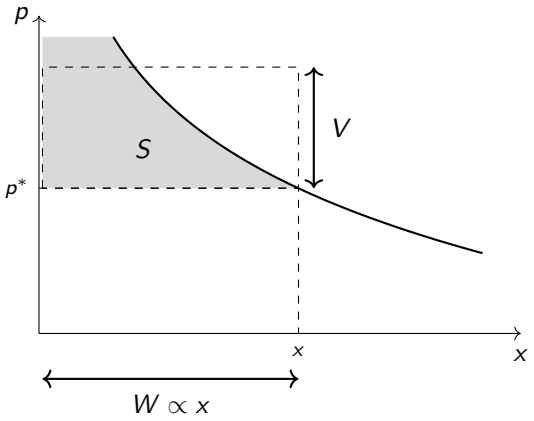
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Real and nominal surplus



- $S = x/\beta$: nominal **per-period consumer surplus**
- $V = 1/\beta$: nominal **per-trip consumer surplus**
  - Published travel-cost estimates report V. Benefit transfer applies V to other sites and years.

Real and nominal surplus



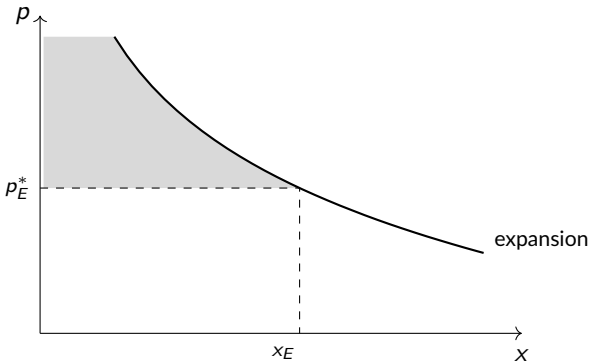
- $S = x/\beta$: nominal **per-period consumer surplus**
- $V = 1/\beta$: nominal **per-trip consumer surplus**
  - Published travel-cost estimates report V. Benefit transfer applies V to other sites and years.
- $W = \dfrac{\lambda(g)}{\lambda(0)}\, S = \dfrac{x}{b\lambda(0)}$: **real surplus**, in mid-cycle dollars

Changes in trips measure changes in real surplus

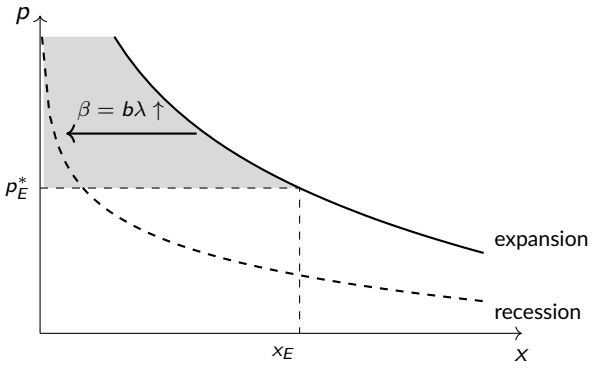
$$\Delta \ln W = \Delta \ln x$$

- Real surplus changes one-for-one with trips.
- Real surplus per trip, $1/(b\lambda(0))$, is constant over the business cycle.
- Nominal surplus per trip is cyclical through λ.

A day hike in a downturn ($s \approx 0.6$)

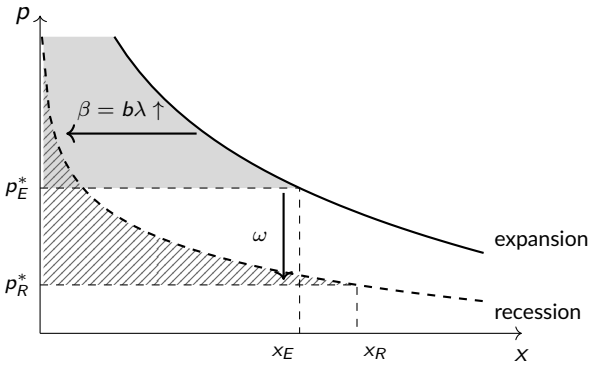


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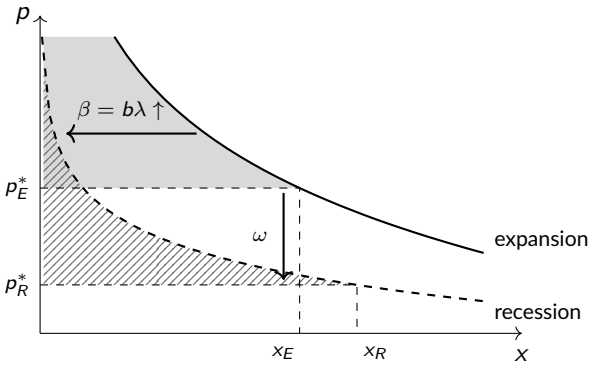
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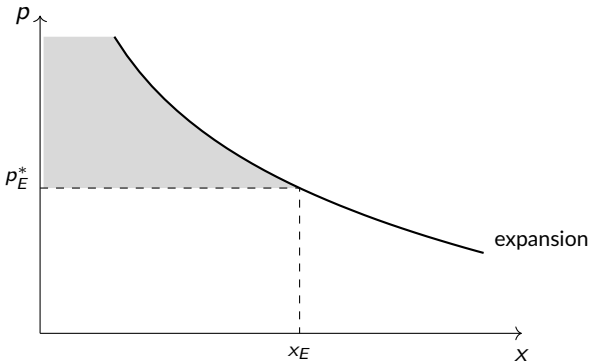
- λ increases, so the demand curve becomes steeper.
- The price of time decreases, so the access price decreases in proportion to s.

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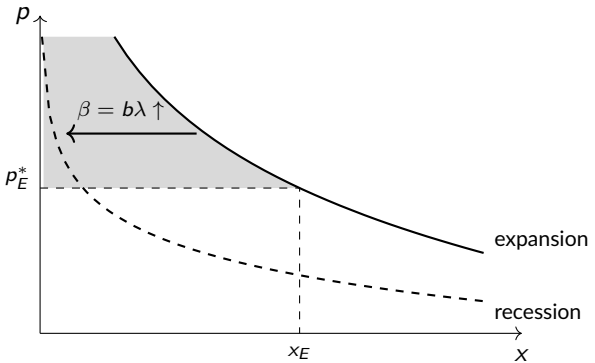


- λ increases, so the demand curve becomes steeper.
- The price of time decreases, so the access price decreases in proportion to s.
- Trips increase, so real surplus increases while nominal surplus decreases.

A ski trip in a downturn ($s \approx 0.4$)

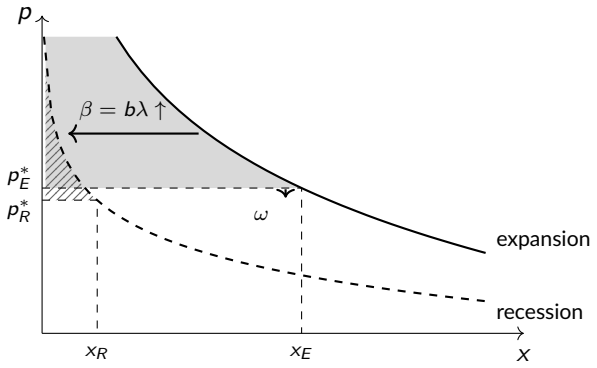


A ski trip in a downturn ($s \approx 0.4$)



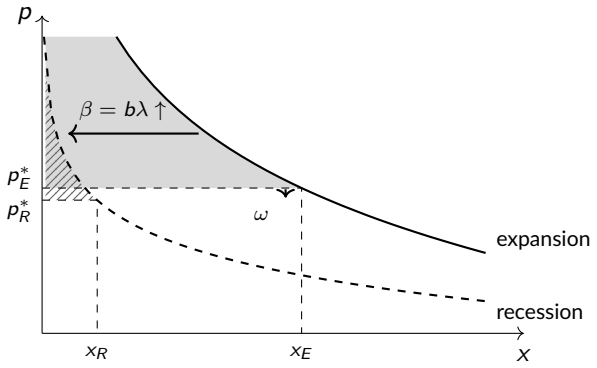
- The same increase in λ makes demand steeper.

A ski trip in a downturn ($s \approx 0.4$)



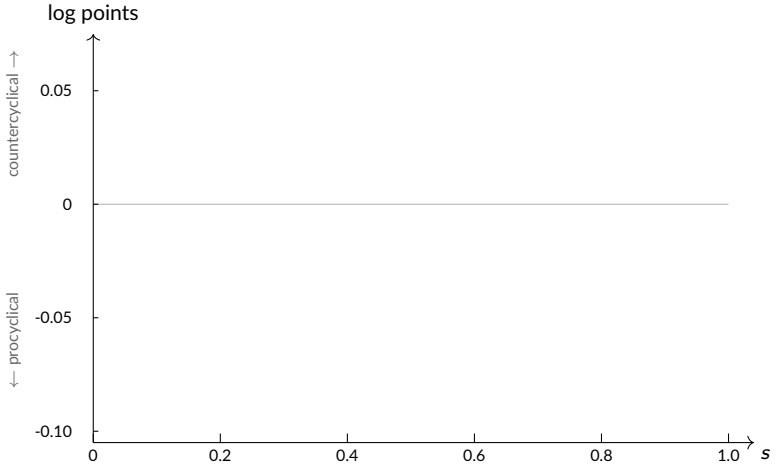
- The same increase in λ makes demand steeper.
- Money costs dominate, so the access price decreases only slightly.

A ski trip in a downturn ($s \approx 0.4$)



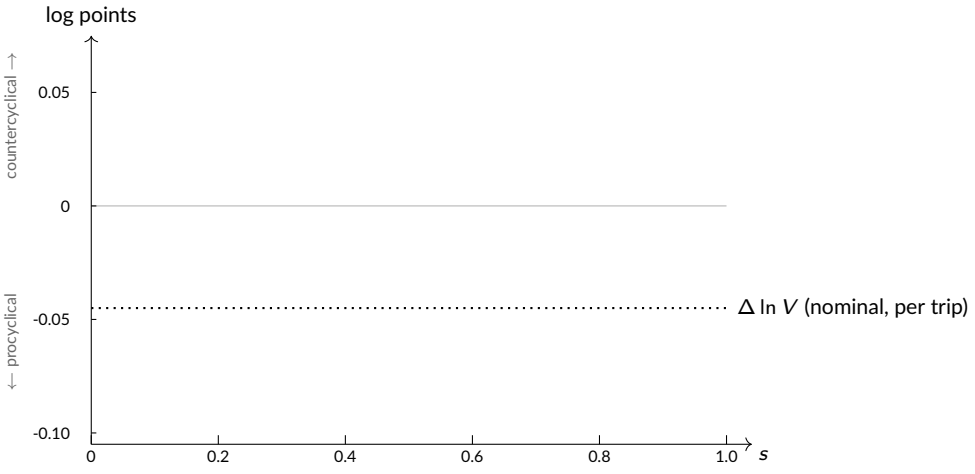
- The same increase in λ makes demand steeper.
- Money costs dominate, so the access price decreases only slightly.
- Trips, real surplus, and nominal surplus all decrease.

Cyclical changes in surplus by travel-time share



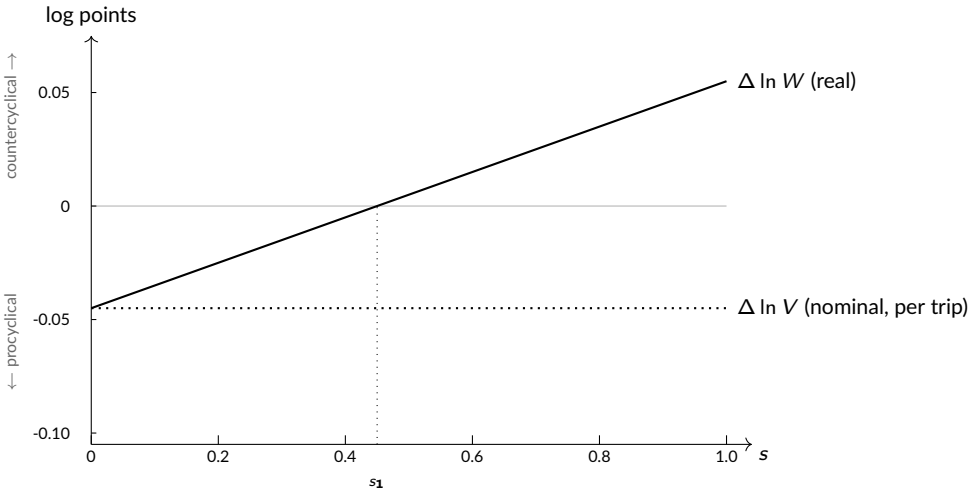
Log change in each surplus measure in a downturn of Great Recession size. Above zero is countercyclical (increases in a downturn). Below zero is procyclical.

Cyclical changes in surplus by travel-time share



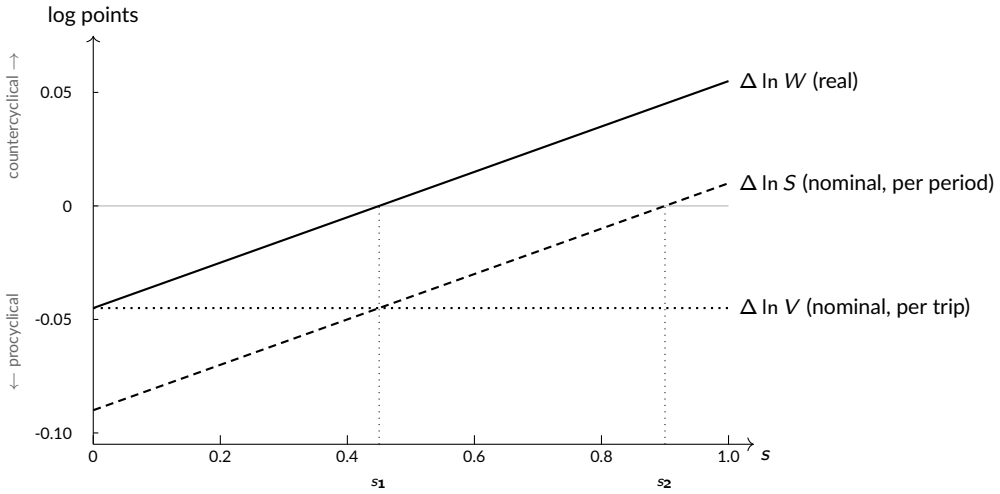
Real surplus per trip is constant, so $\Delta \ln V$ reflects only the change in λ: -4.5 log points at every share.

Cyclical changes in surplus by travel-time share



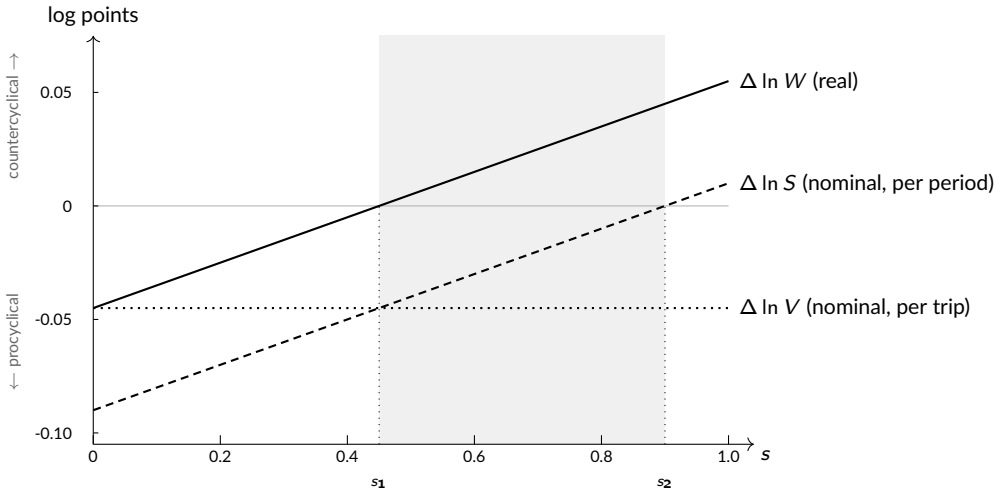
Real surplus increases with the travel-time share and turns positive at $s_1 = 0.45$.

Cyclical changes in surplus by travel-time share



S is below W by the change in λ at every share and turns positive at $s_2 = 0.90$.

Cyclical changes in surplus by travel-time share



In the gray area, real surplus increases in a downturn while every nominal measure decreases.

Cyclicality in National Park visitation?

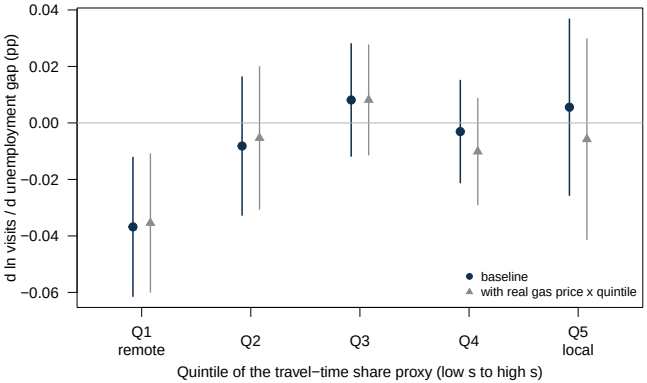
Prediction: the cyclical response of trips increases in s. Trips decrease at money-intensive sites and increase at time-intensive sites.

- **Visits.** NPS monthly recreation visits, 1979–2019. 371 units, 147,266 unit-months.
- **Cycle.** State unemployment gap (LAUS), in percentage points.
- **Share proxy.** Rank of 2020 population within 50 miles of the unit. Yellowstone is low s. Gateway NRA is high s.

$$\ln \text{visits}_{j\tau} = \zeta_{jm} + \kappa_\tau + \sum_\ell \theta_\ell \, g_{h\tau} \times 1[j \in \ell] + \xi_{j\tau}$$

- Unit-by-calendar-month and year-month effects. SEs clustered by state.
- Prediction: θ_ℓ increasing in quintile ℓ, negative at low s.

Visits decrease in downturns at remote NPS units



A 1 pp increase in the state unemployment gap decreases visits by 3.7% (se 1.3) in the most remote quintile. The most local quintile is statistically zero, +0.6% (se 1.6).

Pricing time at the wage introduces a third cyclical bias

Standard practice constructs $\hat{p} = c + kwt$ with $k = 1/3$. Households respond to $p^* = c + \omega t$.

- Both costs scale with distance, so kw is a units error (Randall, 1994). Rescaling the price rescales $\hat{\beta}$.

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$$\hat{V} = \frac{c + kwt}{c + \omega t} \cdot V$$

- The ratio is cyclical: w and ω change by different amounts over the business cycle.

Three cyclical channels in published travel-cost estimates

$$d \ln \hat{V} = \underbrace{\gamma \, d \ln C}_{\text{marginal utility}} + \underbrace{\hat{s} \, d \ln w}_{\text{data-year wage}} - \underbrace{s \, d \ln \omega}_{\text{price of time}}$$

Three cyclical channels in published travel-cost estimates

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- Marginal utility: in every nominal estimate, including stated preference.

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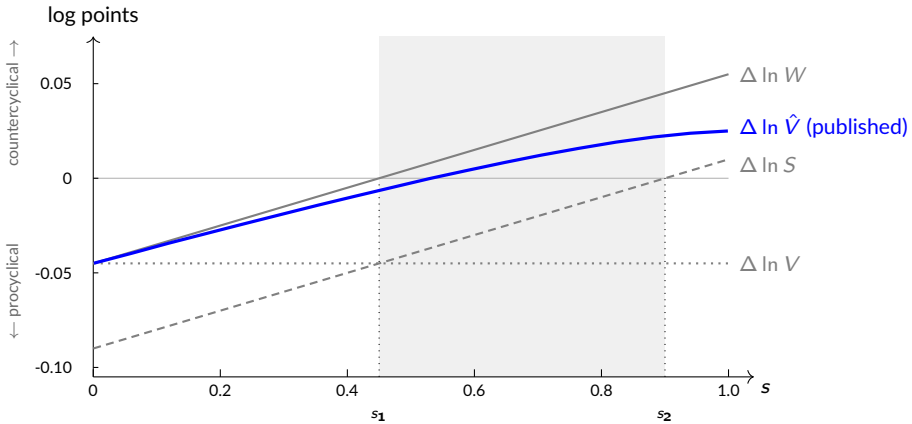
- Marginal utility: in every nominal estimate, including stated preference.
- Data-year wage: only where the researcher priced time at an assumed wage.

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- Marginal utility: in every nominal estimate, including stated preference.
- Data-year wage: only where the researcher priced time at an assumed wage.
- Price of time: in every travel-cost estimate.

Published estimates show a muted effect



- $\Delta \ln \hat{V}$ is close to zero at observed activity shares (0.32 to 0.61).
- Published estimates are less cyclical than any underlying surplus measure.

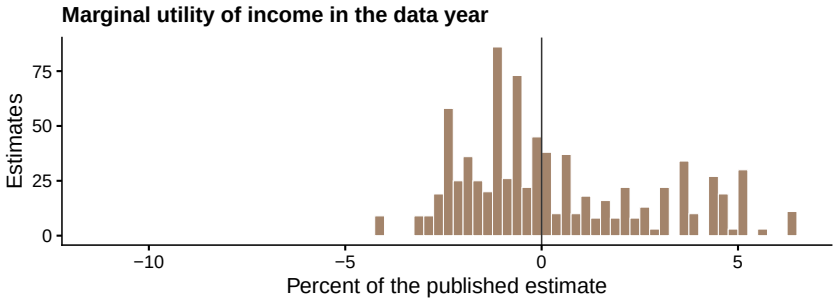
Reassessing published travel-cost estimates

The Recreation Use Values Database (RUVD) compiles 4,023 consumer surplus estimates from 527 publications, 1956–2023, with the state and year of data collection and the assumed price of time.

- Travel time is 0.59 of the per-mile trip price at the mean.
- Activity time shares range from 0.32 for snowmobiling to 0.61 for day hiking, on either side of $s_1 = 0.45$–0.60.

RUVD estimates restated to a mid-cycle labor market

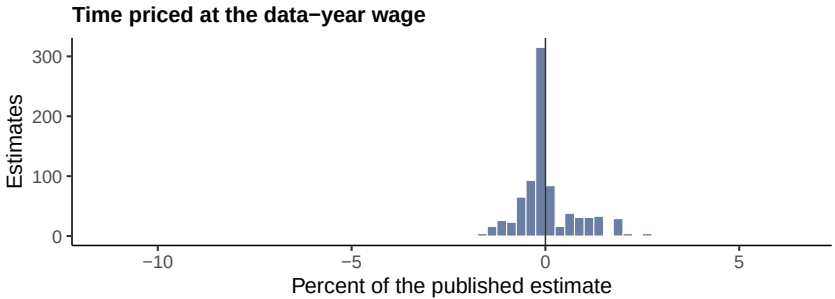
I rescale each published estimate to its state's mid-cycle labor market, channel by channel, using the unemployment gap in its data year.



Marginal utility: -2.3 to $+4.3\%$ (10th to 90th percentile), in every estimate.

RUVD estimates restated to a mid-cycle labor market

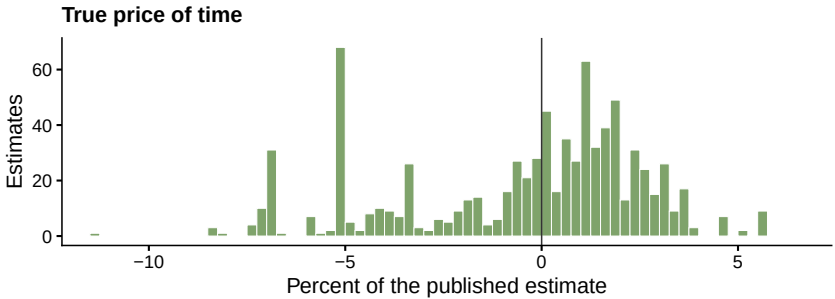
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Data-year wage: -0.6 to $+1.1$%, with a spike at zero for estimates that price no time cost.

RUVD estimates restated to a mid-cycle labor market

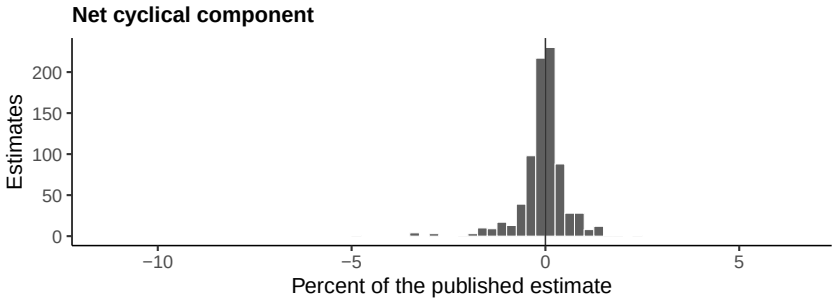
I rescale each published estimate to its state's mid-cycle labor market, channel by channel, using the unemployment gap in its data year.



Price of time: the widest, -5.2 to $+2.9$%, with the opposite sign.

RUVD estimates restated to a mid-cycle labor market

I rescale each published estimate to its state's mid-cycle labor market, channel by channel, using the unemployment gap in its data year.



Net: 80% of estimates are between -0.63 and $+0.49$%. Published estimates are incidentally acyclical.

Deepwater Horizon, revealed preference: the channels offset

Shoreline surveys were conducted in slack labor markets, 2012–13.

- Lost recreational use: $520 million (English et al., 2018).
- Restated to mid-cycle:
  - Marginal utility: $+1.5$ to $+1.8\%$
  - Data-year wage: $+0.6$ to $+0.7\%$
  - Price of time: -2.3 to -2.8%
  - **Net: -0.3%, about $2 million**

The published value is close to what a mid-cycle study would have produced.

Deepwater Horizon, stated preference: one channel, no offset

- Total WTP to avoid DWH damages: $17.2 billion (Bishop et al., 2017).
- Only the marginal utility channel applies.
- The survey was fielded in 2013–14, close to mid-cycle. Restated to mid-cycle, WTP increases by $0.06 billion.

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- Had the survey been fielded in 2010–11, when the damages occurred, the restated WTP would increase by $0.6 billion.
  - More than the entire $520 million shoreline value.

Takeaways

- Trips and real surplus from time-intensive recreation increase in downturns.
- Every nominal estimate decreases in a downturn through the marginal utility channel.
- Published travel-cost estimates are ***incidentally acyclical*** because three channels offset.
- Benefit transfer and damage assessment should restate values to a common point in the business cycle (Johnston et al., 2021; OMB Circular A-4, 2023).
- **Q:** An oil spill occurs in a recession. Which λ should value the lost trips?

The channels offset

For a **one percentage-point increase in the unemployment gap**, at central parameters:

marginal utility	$+1.1$ percent
data-year wage	$+0.25$ percent
price of time	-1.5 percent
net	-0.2 percent (median)

- Stated-preference estimates have no constructed price of time, so nothing offsets the marginal utility channel.

Why not just increase k?

- Revealed-preference estimates place the value of travel time near the full price of time, not a third of the wage (Fezzi et al., 2014; Goldszmidt et al., 2020; Lloyd-Smith et al., 2019; Wichman and Cunningham, 2023).
- Increasing k shrinks the cyclical error but no constant k eliminates it: the wage omits the employment margin.

The wage convention passes through to published values

- Within publications, the assumed wage fraction passes through to the published estimate with a slope of 1.19 (s.e. 0.18), across 367 pairs.
- A slope of one is what the ratio $\hat{V}/V$ predicts: the analyst's choice of k rescales the estimate one-for-one.

Calibration parameters

Per point of state unemployment gap:

	central	range
wage, ϕ_w	1.0 percent	0.5–2.0
consumption, ϕ_c	0.75 percent	0.5–1.0
price of time, ϕ_ω	2.5 percent	1.5–4.0
curvature, γ	1.5	1–2

A four-point gap reproduces the Great Recession magnitudes.

Level regressions

Log published estimate on the state unemployment gap in the data year, decade fixed effects, clustered by publication:

- Travel cost: -0.7 percent per gap point (s.e. 4.7), consistent with the offsetting channels.
- Stated preference: $+4.5$ percent (s.e. 3.4), against a predicted -0.5 to -2.0.
- These regressions are too imprecise to detect responses of one to two percent per point.

The threshold arithmetic

$$s_1 = \frac{\gamma \, \Delta \ln C}{\Delta \ln \omega} = \frac{0.045}{0.075 \text{ to } 0.10} = 0.45 \text{ to } 0.60$$

$$s_2 = s_1 \, \frac{1 + |\varepsilon|}{|\varepsilon|} = 2 \, s_1 \quad \text{at } |\varepsilon| = 1$$

Use increases in a downturn once travel time is roughly half of the trip price. Full-cost shares at the full wage: day hiking 0.61, general day use 0.49, downhill skiing 0.38, snowmobiling 0.32.